



RRDGNN: Relational reflective disentangled graph neural network for entity alignment

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Received: 8 April 2025 / Accepted: 21 May 2025 / Published online: 28 May 2025
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Abstract

The objective of entity alignment (EA) is to automatically discover entities across different knowledge graphs (KGs) that refer to the same real-world object, which is a crucial step for knowledge fusion. Graph neural network based methods currently dominate the EA landscape. However, the entity neighborhood often contains information about various aspects. Most of these methods treat the neighborhood of an entity as a whole, aggregating all its neighbors into a single, static representation. This can introduce noise and lead to incorrect alignment results due to the heterogeneous neighborhood structures of aligned entities. In this paper, we propose a novel relational reflective disentangled graph neural network (RRDGNN) for EA, which learns disentangled representations for different aspects of entities, thereby alleviating the prevalent issue of neighborhood heterogeneity in EA. Specifically, RRDGNN designs a relational reflective neighborhood routing mechanism for learning disentangled, dynamic entity representations at both micro and macro levels. An alignment-aware self-attention strategy is then employed to assign adaptive weights to different entity aspects. Finally, multi-aspect seed generators are co-trained in a semi-supervised learning framework. They favor entities with specific sparsity, which helps to refine the seed selection process and generate high-quality alignment seeds. Extensive experiments on five real-world datasets, along with detailed ablation studies and analyses, demonstrate that RRDGNN consistently outperforms existing EA methods.

Keywords Knowledge graph · Entity alignment · Disentangled representation learning · Graph neural network

1 Introduction

Knowledge graphs (KGs), which support AI applications such as information retrieval, knowledge base question answering, and recommender systems, have developed rapidly in recent years. Entity alignment (EA) enables the automatic discovery of equivalent entities across different KGs and has become a pivotal technique for knowledge fusion, enhancing knowledge coverage. Consequently, it has garnered significant interest from researchers. Among the various EA methods, graph neural networks (GNNs) based approaches (Wang et al. 2018; Cao et al. 2019; Sun et al. 2020; Akhtar et al. 2022; Shi et al. 2023) have yielded encouraging results

and have become the current mainstream. Different GNN variants are designed to encode neighborhood information into entity embeddings, allowing for the capture of structural similarities between equivalent entities.

Nevertheless, most prior works in this field are founded upon a significant assumption: that entities and their counterparts in other KGs possess analogous neighborhood structures. Following GNN encoding, the equivalent entities will generate similar embeddings. In practice, different KGs are built by different institutions using different data sources for specific purposes. They often emphasize different aspects of entities, with some aspects remaining consistent across KGs while others vary. The entity *Stephen Curry*, illustrated in Fig. 1, serves as an intuitive example. In an open domain KG (see Fig. 1(a)), the entity is associated with entities related to his family, career, place of birth, and nationality. In a basketball-related vertical domain KG (see Fig. 1(b)), the neighbors of this entity primarily include other basketball players, his career information, and the awards he has won. This phenomenon is commonly observed in entity alignment, known as **neighborhood heterogeneity**, where the neighbor-

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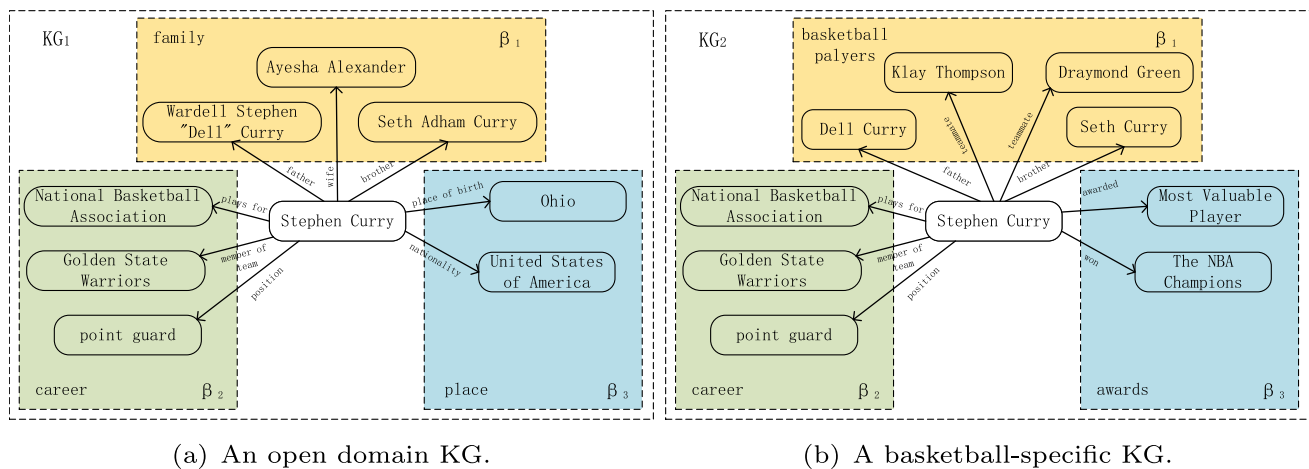


Fig. 1 An example used to illustrate the neighborhood heterogeneity of equivalent entities. Fig. 1(a) presents a local subgraph derived from an open domain KG, and Fig. 1(b) displays a local subgraph from a

basketball-related KG. Different colors indicate different aspects. It is clear that different KGs do not focus on entities from the same perspective

hood structures of equivalent entities in different KGs can vary significantly. Furthermore, Sun et al. (2020) conducted statistical analyses to determine that the proportion of aligned entity pairs with heterogeneous neighborhoods exceeds 86% in all three sub-datasets of DBP15K (Sun et al. 2017), the benchmark dataset for EA. Heterogeneous neighborhood structures will induce the GNNs to generate different representations for equivalent entities. Some previous works attempt to mitigate neighborhood heterogeneity, for instance, AliNet (Sun et al. 2020) tries to aggregate both one-hop and multi-hop neighbors, aiming to improve the overlap among the neighborhoods of aligned entities. RSA (Akhtar et al. 2022) exploits bidirectional neighborhood path information and relational semantics contexts to enhance entity representations. BNGNN (Shi et al. 2023) accounts for both entity structure similarity and attribute similarity, and reinforces the interaction between these two types of features in each aggregation layer. However, the results of these methods are not particularly satisfactory. They are limited by the inherent constraints of traditional GNNs, i.e., treating the local neighborhood of *Stephen Curry* as a whole and aggregating all its neighbors into a single, static representation. This encoding strategy couples information from various aspects of an entity and fails to explicitly distinguish the contributions of each aspect, inevitably introducing noise and resulting in incorrect alignment results.

Recently, researchers have put forth the concept of disentangled graph representation learning (Ma et al. 2019; Liu et al. 2020; Zheng et al. 2024). It enables the separation of distinct factors of variation behind the graph data. Inspired by this, the primary objective of this research is to address the neighborhood heterogeneity problem by learning **disentangled entity representations**. Specifically, the

neighborhoods of entities are automatically divided into multiple semantically different aspects. A corresponding embedding is learned for each aspect, which can characterize the entity from this disentangled aspect. During the alignment process, higher weights are assigned to more similar aspects while lower weights are allocated to less relevant ones. Referring to the example in Fig. 1, the neighborhoods of the entities to be aligned in the two KGs are divided into three parts. The green sub-neighborhoods are similar, the blue sub-neighborhoods are dissimilar, and the yellow sub-neighborhoods are partially similar, as both Curry's father and brother are also basketball players. Therefore, we expect $\beta_2 > \beta_1 > \beta_3$ to hold. The significance of disentangling KG structure can be quantitatively analyzed. From the perspective of graph structure, the similarity between two entities (e_1, e_2) can be defined as $s(e_1, e_2) = \frac{|N_{e_1} \cap N_{e_2}|}{|N_{e_1} \cup N_{e_2}|}$, where N_e denotes the set of neighbors of entity e . If the neighborhood of the entity *Stephen Curry* is considered as a whole, $s(e_1, e_2) = \frac{5}{12}$. If its neighborhood is divided into three parts and β_1, β_2 , and β_3 are equal to 0.3, 0.6, and 0.1 respectively, then $s(e_1, e_2) = \alpha_1 \frac{2}{5} + \alpha_2 \frac{3}{3} + \alpha_3 \frac{0}{4} = 0.72 > \frac{5}{12}$. β_1, β_2 , and β_3 are trainable parameters that can be optimized to meet this condition. Although the analysis differs somewhat from the actual alignment process, it still demonstrates the reasonableness and effectiveness of disentangled representations.

Unfortunately, disentangling graph structures in EA tasks is non-trivial, presenting three main challenges. (1) How to fully exploit semantic information of relations to guide the partition of the neighborhood structure: Most existing neighborhood routing algorithms are designed for homogeneous graphs, while KGs are typical multi-relational graphs. In Fig. 1(a), the neighboring entities connected by the relations *father* and *brother* should be divided into the same

sub-neighborhood, as they both pertain to the entity's *family* aspect. (2) How to assign weights to different aspects adaptively: As illustrated in Fig. 1, different aspects do not contribute equally to the alignment results. Due to different focuses, some aspects may even introduce noise. Therefore, it is essential to dynamically assign appropriate weights to different aspects when aligning entities. (3) How to set different numbers of aspects for different informative entities: Current disentangled graph representation learning methods uniformly divide the neighborhoods of all nodes into a fixed number. However, entities in KG contain varying amounts of information. Intuitively, the neighborhoods of high-degree nodes should be divided into more subsets, while those of low-degree nodes should be divided into fewer subsets.

To deal with the above problems, (1) firstly, we propose a novel relational reflective disentangled graph neural network (RRDGNN). It can explicitly utilize the relation information to guide the partition of entity neighborhoods via a relational reflective neighborhood routing mechanism. The disentangled representations of different entity aspects are achieved at both micro and macro levels. Specifically, in micro-disentanglement, entity embeddings reflected by connected relations are clustered in multiple rounds, yielding a corresponding representation for each aspect. In terms of macro-disentanglement, we impose mutual information regularization on representations of the disentangled aspects. This constraint promotes independence among the representations of different aspects and enhances the diversity of the information they contain. (2) Secondly, we design an alignment-aware self-attention. For a given entity pair, attention weights are dynamically assigned to different aspects. The similarity of embeddings in each aspect is then calculated, and the similarities of various aspects are combined to produce the final alignment result. (3) Finally, we devise a semi-supervised learning framework with multi-aspect seed generators. A **seed generator** is an instance of RRDGNN that iteratively labels potentially aligned entity pairs and incorporates them into the training data. We provide a simple definition of entity sparsity. Then multiple seed generators with distinct aspect values are co-trained in this framework, and each generator adaptively selects seeds with specific sparsity according to its aspect value. We also experimentally verify that the degree distribution of seeds varies among different generators. Consequently, the labeling outcomes for entities with different sparsities are determined by different generators. Furthermore, the collaboration among multiple generators helps to refine the seed selection process, leading to more robust alignment performance.

In summary, the main contributions of this paper can be summarized as follows:

- We propose a novel model RRDGNN for entity alignment. It exploits the relational reflective neighborhood

routing mechanism to learn disentangled, dynamic entity representations at both micro and macro levels. To the best of our knowledge, this is the first application of the graph disentanglement concept to the EA task.

- We design an alignment-aware self-attention strategy that can adaptively assign appropriate weights to different aspects when calculating the similarity of entity pairs. Based on this, the widespread neighborhood heterogeneity phenomenon in EA is alleviated.
- We propose multi-aspect seed generators. In a semi-supervised learning framework, these generators with distinct aspect values can be co-trained to adaptively select new entity pair seeds with specific sparsity, thereby augmenting the training data.
- We conduct extensive experiments on several real-world datasets, and the results demonstrate that RRDGNN not only consistently outperforms existing EA methods, but also exhibits strong robustness. Additionally, subsequent ablation studies validate the effectiveness of key components.

The remainder of this paper is organized as follows. Section 2 introduces recent related work on entity alignment and disentangled graph representation learning. Section 3 formally defines the EA problem and Section 4 describes our method RRDGNN in detail. Section 5 presents a comprehensive evaluation of the proposed method. Section 6 summarizes the paper and outlines potential avenues for future research.

2 Related work

Early entity alignment was accomplished primarily through similarity calculation and relational reasoning. However, these methods require extensive manual annotation and the design of complex rules. The advancement of knowledge representation learning technologies has led to the emergence of embedding-based methods, which present a new solution for EA. They embed entities and relations in KGs into a continuous low-dimensional space and ascertain whether entities are aligned by evaluating the similarity between entity embeddings. Depending on the encoding model used, embedding-based entity alignment can be roughly divided into two categories: translation-based and GNN-based methods. In addition to the relation triples, some additional information is often used to assist in aligning entities. In this section, we review related work on *translation-based*, *GNN-based*, and *additional information-based* EA, along with *disentangled graph representation learning*.

2.1 Translation-based EA

Among various translation models, TransE (Bordes et al. 2013) is the most widely used one for EA. For a triple (h, r, t) , TransE considers the relation embedding as the translation vector from the head entity embedding to the tail entity embedding. MTransE (Chen et al. 2017) first encodes entities and relations within a single KG into separate embedding spaces, then proposes an alignment module with three different techniques to construct transformations across different spaces. BootEA (Sun et al. 2018) proposes truncated uniform negative sampling to select similar entities as negative samples and expand positive samples through parameter exchange. In addition, it adopts a bootstrap strategy that iteratively labels predicted aligned entity pairs as new alignment seeds. TransEdge (Sun et al. 2019b) is an edge-centric encoding model that treats specific head and tail entities in each triple as the context of the relation, thereby learning contextualized relation representations. ESEA (Jiang et al. 2022) designs a hybrid framework that allows for the combination of the strengths of symbolic-based models and embedding-based models to achieve better EA results. However, translation-based methods can only learn local features of entities and tend to produce similar embeddings for entities connected with many-to-one or one-to-many relations.

2.2 GNN-based EA

Graph neural networks obtain node representations by aggregating information from neighbors. GCN-Align (Wang et al. 2018) is the first attempt to utilize graph convolutional networks for EA. It obtains entity representations via neighborhood aggregation and infers alignment results based on entity similarity in a unified embedding space. To alleviate the structural differences between different KGs, AliNet (Sun et al. 2020) expands the neighborhood overlap of corresponding entities by aggregating distant neighbors. In each layer of the network, it applies a gating mechanism to combine representations of direct and distant neighbors. However, relation information has been neglected in the above methods, and some studies (Mao et al. 2020a, b; Akhtar et al. 2022; Zhang et al. 2024; Akhtar et al. 2025) have attempted to explore the role of relations in EA tasks. MRAEA (Mao et al. 2020a) considers the effects of the meta-semantics and directions of relations, and designs relation-aware self-attention that can focus on the neighbors of entities as well as the relations between them when updating entity representations. Observing that the standard linear transformation in GNN is detrimental to EA performance, RREA (Mao et al. 2020b) proposes two key metrics to achieve an ideal transformation operation. Furthermore, it learns relation-specific entity representations via relation-reflective transformation. RSA (Akhtar et al. 2022) computes the centrality of entities and

relations and selects the most informative relational semantic context to enhance entity representations. RHGNN (Zhang et al. 2024) uses a heterogeneous graph neural network to model entities and relations simultaneously and promotes the interaction between entities and relations through an iterative fusion strategy. AKR (Akhtar et al. 2025) is an abductive knowledge reasoning model for entity alignment, which captures complex relational semantics via KG reasoning and enriches entity context through graph centrality calculation. Dual-AMN (Mao et al. 2021) contains a relation attention layer and a proxy matching attention layer. The former can capture neighborhood information within a KG, and the latter can model across-KG relations. In addition, it proposes a normalized hard sample mining loss for more effective negative sampling. Similarly, DuGa-DIT (Xie et al. 2021) also considers intra-KG and cross-KG attention and dynamically updates the cross-KG score matrix by iterative training. GALA (Zhang et al. 2023) extracts global features from different KGs, which can improve the alignment probability of entities far away from pre-aligned entity pairs. Furthermore, it proposes a semi-supervised learning strategy to avoid using fixed entity labels. SEAGAN (Qian and Pan 2023b) incorporates two effective components: one for selecting reliable new entity pairs through a variety-aware GAN, and the other for preventing overfitting with an online learning algorithm. HOLI-GNN (Chen et al. 2024) focuses on learning high-order structural information of entities for EA. To reduce the negative impact of over-smoothing, it develops a local inflation mechanism that can enlarge entities' features. SPSEA (Yan et al. 2024) propagates alignment information from alignment seeds to their direct neighbors, alleviating the mislabeling problem in semi-supervised entity alignment.

However, the existing GNN-based methods treat the neighborhood of entities as a whole, and thus can only learn single, static entity representations. In this paper, we apply the idea of graph disentanglement to EA and propose a novel method, RRDGNN. By learning disentangled, dynamic entity representations at both micro and macro levels, it alleviates heterogeneous neighborhood structures and demonstrates enhanced robustness and explainability.

2.3 Additional information-based EA

In addition to the structural information present in relation triples, some studies enhance EA results by introducing additional entity information. JAPE (Sun et al. 2017) consists of two modules: structure embedding and attribute embedding, which are used to model relational structure and attribute correlation, respectively. By combining these two modules, JAPE jointly embeds entities from disparate KGs into a unified space. Subsequent studies (Zhong et al. 2022; Tam et al. 2022) also refer to the learning framework of JAPE, which includes two embedding modules. SDEA (Zhong et al. 2022)

contains a BiGRU-based relation embedding module and a transformer-based attribute embedding module. EMGCN (Tam et al. 2022) adopts a multi-order GCN for modeling relation correlation and machine translation for capturing attribute correlation. Different features are combined via late fusion to obtain more informative entity embeddings. Different from the above methods, BNGNN (Shi et al. 2023) models both relation triples and attribute triples with a unified graph. It can fully fuse these two types of information in each layer. Besides attribute information of entities, literal descriptions (Chen et al. 2018), position importance (Qian and Pan 2023a), images (Su et al. 2023; Wang et al. 2024; Li et al. 2024), and public text corpora (Chen et al. 2021) are also used to enhance the performance of EA. To obtain more comprehensive entity representations, SCMEA (Zhou et al. 2024) considers various types of entity information and integrates them into a unified embedding space. It also introduces an enhanced bidirectional contrastive learning loss to mitigate the hubness problem. SCSA (Li et al. 2025) employs Hypergraph Neural Networks to capture neighborhood relational similarity, thereby enhancing entity representations. During iterative learning, it integrates diverse auxiliary information to improve the quality of pseudo-labeled data. In this

paper, our method RRDGNN does not rely on such external information. However, we have developed a variant of RRDGNN that incorporates pre-trained word embeddings of entity names. This allows for a comparison with the aforementioned methods in the experiments.

Table 1 presents a comprehensive summary of related EA methods in terms of embedding model, additional information, and learning strategy.

2.4 Disentangled graph representation learning

Graph neural networks are powerful machine learning models designed to handle graph-structured data, where the inputs are represented as graphs. Numerous GNN variants, such as graph convolutional network (GCN) (Kipf and Welling 2017), gated graph neural network (GGNN) (Li et al. 2016), graph attention network (GAT) (Veličković et al. 2018), relational graph convolutional networks (R-GCNs) (Schlichtkrull et al. 2018), and hyperbolic graph convolutional networks (HGCN) (Chami et al. 2019) have attracted considerable attention recently. Despite promising results in many tasks, most existing GNNs treat the central node's neighborhood as a whole and ignore semantic differences

Table 1 Summary of related EA methods

Methods	Embedding model	Additional information	Learning strategy
MTransE (Chen et al. 2017)	TransE	-	Supervised
BootEA (Sun et al. 2018)	TransE	-	Semi-supervised
TransEdge (Sun et al. 2019b)	TransEdge	-	Semi-supervised
ESEA (Jiang et al. 2022)	TransE & Rule embedding	Rule & Literal information	Supervised
GCN-Align (Wang et al. 2018)	GCN	Attributes	Supervised
AliNet (Sun et al. 2020)	GNN	-	Supervised
MRAEA (Mao et al. 2020a)	GNN	-	Semi-Supervised
RREA (Mao et al. 2020b)	GNN	-	Semi-Supervised
RSA (Akhtar et al. 2022)	GNN	Entity names	Supervised
RHGNN (Zhang et al. 2024)	GNN	Entity names	Supervised
AKR (Akhtar et al. 2025)	GNN	Entity names	Semi-supervised
Dual-AMN (Mao et al. 2021)	GNN	-	Semi-supervised
DuGa-DIT (Xie et al. 2021)	GNN	Entity names	Semi-supervised
GALA (Zhang et al. 2023)	GNN	-	Semi-supervised
SEAGAN (Qian and Pan 2023b)	GNN	-	Semi-supervised
HOLI-GNN (Chen et al. 2024)	GNN	Entity names	Semi-supervised
SPSEA (Yan et al. 2024)	GNN	-	Semi-supervised
JAPE (Sun et al. 2017)	TransE	Attributes	Supervised
SDEA (Zhong et al. 2022)	BiGRU & BERT	Attributes & Entity names	Supervised
EMGCN (Tam et al. 2022)	GNN	Attributes & Entity names	Supervised
BNGNN (Shi et al. 2023)	GNN	Attributes & Entity names	Semi-supervised
SCMEA (Zhou et al. 2024)	GNN	Attributes & Entity names	Semi-supervised
SCSA (Li et al. 2025)	GNN	Attributes & Entity names	Semi-supervised

among different parts within the neighborhood, resulting in learned representations with weak robustness and explainability.

To address this issue, DisenGCN (Ma et al. 2019) assumes that the edge connections between nodes are determined by several inherent factors behind the data, which shape the nodes from different aspects. It is the first work to focus on graph disentanglement from the node level. However, DisenGCN is deficient in its failure to consider macro-separability. ADGCN (Zheng et al. 2024) and IPGDN (Liu et al. 2020) further achieve macro-disentanglement by leveraging adversarial learning and the Hilbert-Schmidt independence criterion, respectively, to ensure the independence of disentangled representations. LGD-GCN (Guo et al. 2024) enhances intra-factor consistency and inter-factor diversity through global message passing and local neighborhood routing for more effective graph disentanglement. In addition to node classification, the idea of disentangled graph representation learning has been used for many graph-based tasks, such as recommender systems (Mu et al. 2021; Xu et al. 2024), knowledge completion (Wu et al. 2021), and pedestrian trajectory prediction (Bae and Jeon 2021). KGs are multi-relational heterogeneous graphs in which the relations have rich semantics. The objective of this research is entity alignment, and we propose a novel relational reflective disentangled graph neural network for this task. It explicitly leverages the relation information to guide the partition of entity neighborhoods. Moreover, the above approaches introduce additional trainable parameters as transformation matrices to project entity embeddings into the corresponding aspect's subspace, while our method requires no extra parameters.

3 Preliminaries

In this paper, a knowledge graph is denoted as $KG = (E, R, T)$, where E and R are the sets of entities and relations. $T = \{(h, r, t) \in E \times R \times E\}$ is the set of relation triples, where h is a head entity, t is a tail entity and r is a directed relation connecting them.

We investigate the case of entity alignment between two KGs, i.e., the source KG and the target KG. Furthermore, we focus only on one-to-one entity alignment. In other words, for an entity from the source KG, there is one and only one counterpart entity in the target KG. The EA problem is formally defined as follows:

Let $KG_1 = (E_1, R_1, T_1)$ and $KG_2 = (E_2, R_2, T_2)$ be two heterogeneous KGs to be aligned. A set of pre-aligned entity pairs (also known as alignment seeds) $P = \{(e_{i1}, e_{i2}) | e_{i1} \in E_1, e_{i2} \in E_2, e_{i1} \equiv e_{i2}\}_{i=1}^{|P|}$ is provided as training data, where $\equiv$ indicates that entities e_{i1} and e_{i2} point to the same object in the real world. The goal of EA is to find more aligned entities in E_1 and E_2 , excluding the alignment seeds P .

4 Our method

This section commences with an overview of our method. Thereafter, the implementation details of each component are described thoroughly.

4.1 Overall architecture

As illustrated in Fig. 2, our method takes two KGs (KG1 and KG2) and a few pre-aligned entity pairs as input, and

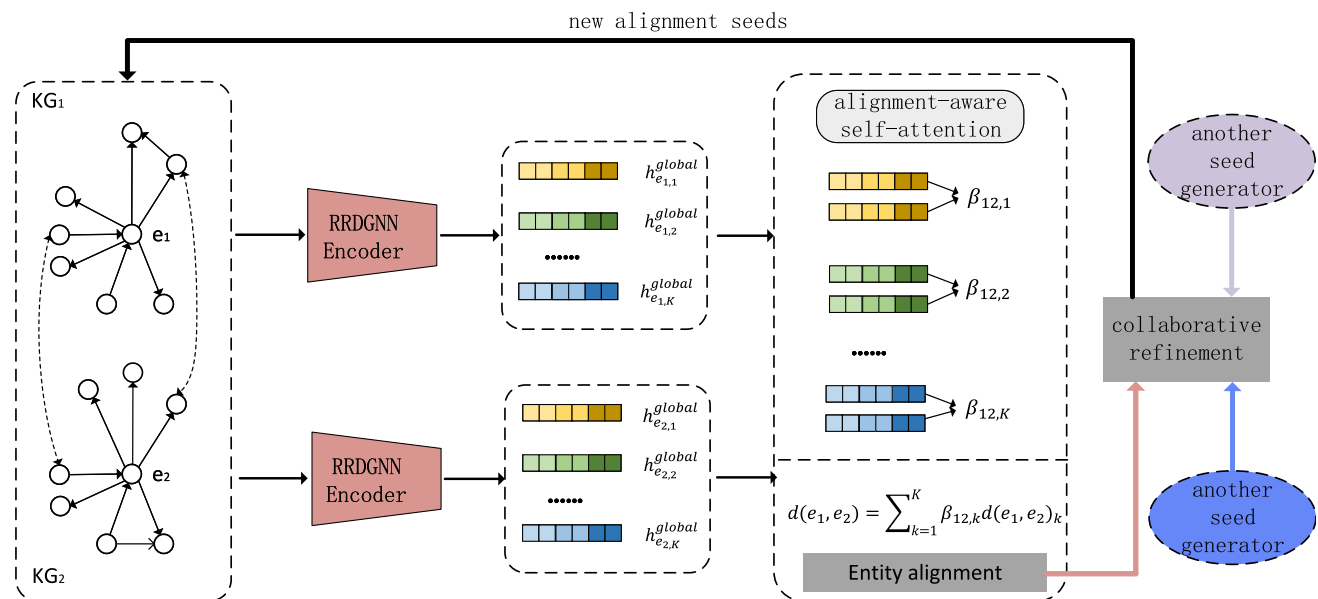


Fig. 2 Architecture of our method. e_1 and e_2 are entities to be aligned. Pre-aligned entities are connected with double-arrow dashed curves

the output is the predicted newly aligned entity pairs. This alignment process is enabled by the following four steps. (1) First, a disentangled graph encoder is proposed to implement micro-disentanglement. In each layer, based on the relational reflective neighborhood routing mechanism, entities' information about different aspects is encoded into the corresponding disentangled embeddings. Multiple layers are stacked to capture higher-order neighborhood features. (2) Subsequently, mutual information regularization is applied to the final disentangled entity representations to achieve macro-disentanglement. This constraint serves to reduce the correlation between embeddings of different aspects. The encoding process of RRDGNN is shown in Fig. 3. (3) Next, entity alignment is conducted using alignment-aware self-attention. Different aspects are assigned adaptive attention weights, and the similarities of different aspect embeddings are combined to obtain comprehensive alignment results. (4) Finally, in a semi-supervised learning framework, multiple seed generators with different aspect values are trained simultaneously. The generators can iteratively select new entity pair seeds with specific sparsity. In this way, the data augmentation process can be refined through the collaboration of multiple generators to yield high-quality alignment seeds.

4.2 Micro-disentanglement with relational reflective neighborhood routing

The objective of micro-disentanglement is to obtain disentangled entity representations by encoding features from various aspects. Some previous studies (Ma et al. 2019; Liu et al. 2020; Zheng et al. 2024) have employed neighborhood rout-

ing algorithms to automatically segment the central node's neighbors into different subsets. However, these methods are not directly applicable to KG representation learning because they overlook the inherent relation information, which can explain why two entities are connected. Therefore, we propose a relational reflective neighborhood routing mechanism that explicitly leverages the relation semantics to guide the partition of entity neighborhoods.

In contrast to previous EA methods that parameterize entities with a holistic representation, we use K independent embeddings to jointly represent the entity e_i :

$$\mathbf{h}_{e_i} = [\mathbf{h}_{e_i,1}, \mathbf{h}_{e_i,2}, \dots, \mathbf{h}_{e_i,K}] \in \mathbb{R}^{Kd}, \quad (1)$$

where $\mathbf{h}_{e_i,k}$ is the disentangled representation for describing the k -th aspect. Analogously, the representation of the relation r_m also consists of K parts: $\mathbf{h}_{r_m} = [\mathbf{h}_{r_m,1}, \mathbf{h}_{r_m,2}, \dots, \mathbf{h}_{r_m,K}] \in \mathbb{R}^{Kd}$. Embeddings of entities and relations for different aspects are randomly initialized at the beginning of training.

Inspired by Mao et al. (2020b), we establish the following relational reflective transformation matrix for relation r_m , which can project entity embeddings to the subspace corresponding to the k -th aspect:

$$\mathbf{M}_{r_m,k} = \mathbf{I} - 2\mathbf{h}_{r_m,k}\mathbf{h}_{r_m,k}^T, \quad (2)$$

where $\mathbf{I}$ is an identity matrix. When the embedding $\mathbf{h}_{r_m,k}$ is normalized to $\|\mathbf{h}_{r_m,k}\|_2 = 1$, $\mathbf{M}_{r_m,k}$ will be an orthogonal matrix. Transformation matrices with orthogonality can retain norms and relative distance of entity embeddings.

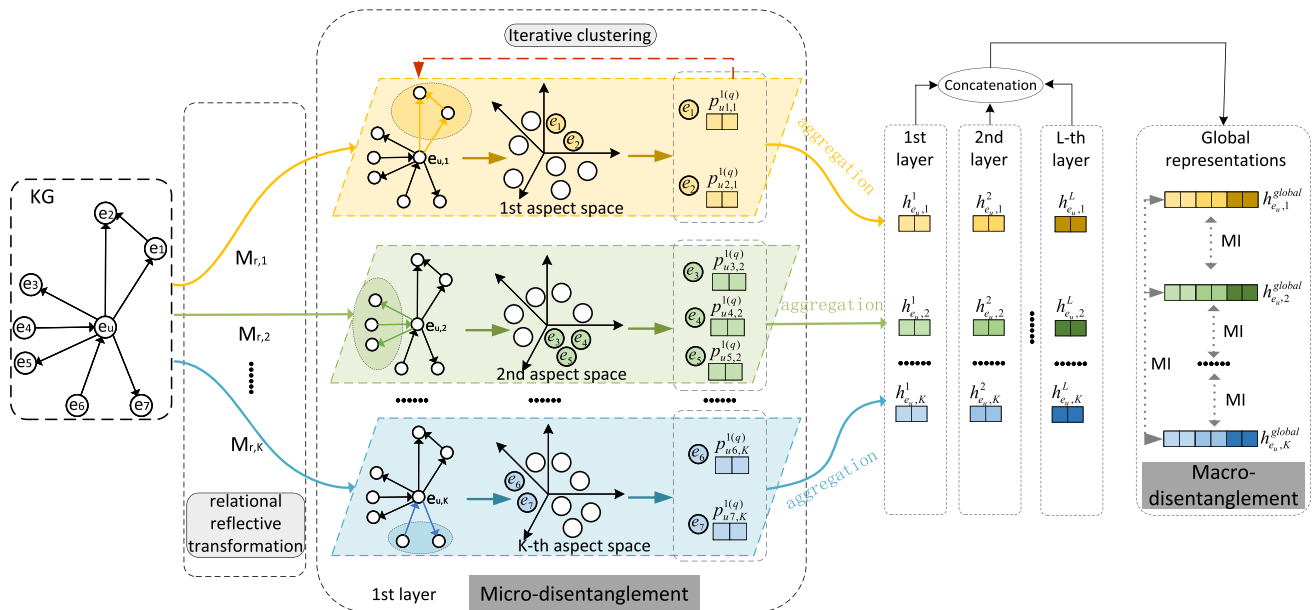


Fig. 3 The encoding process of RRDGNN

Next, given a central entity e_i and all the triples involving it, each RRDGNN layer attempts to learn disentangled representations of e_i oriented to different aspects. To achieve this goal, the following Q-round clustering is iteratively executed:

$$\mathbf{c}_{e_i,k}^{l(q)} = \mathbf{h}_{e_i,k}^l + \sum_{(r_m, e_j) \in N_{e_i}} p_{ij,k}^{l(q-1)} \mathbf{M}_{r_m,k} \mathbf{h}_{e_j,k}^l, \quad (3)$$

$$p_{ij,k}^{l(q)} = \frac{\exp(\text{sim}(\mathbf{c}_{e_i,k}^{l(q)}, \mathbf{M}_{r_m,k} \mathbf{h}_{e_j,k}^l) / \tau)}{\sum_{k'=1}^K \exp(\text{sim}(\mathbf{c}_{e_i,k}^{l(q)}, \mathbf{M}_{r_m,k'} \mathbf{h}_{e_j,k'}^l) / \tau)}, \quad (4)$$

$$\text{sim}(\mathbf{c}_{e_i,k}^{l(q)}, \mathbf{M}_{r_m,k} \mathbf{h}_{e_j,k}^l) = -\|\mathbf{c}_{e_i,k}^{l(q)} - \mathbf{M}_{r_m,k} \mathbf{h}_{e_j,k}^l\|_{L_1}, \quad (5)$$

where q is the current iteration number, N_{e_i} is the set of entity-relation pairs neighboring e_i , $\mathbf{h}_{e_i,k}^l$ and $\mathbf{h}_{e_j,k}^l$ are the input hidden representations of entities e_i and e_j in the k -th aspect at the l -th layer, and τ is a hyperparameter used as a temperature scale. $\mathbf{c}_{e_i,k}^{l(q)}$ can be considered as the center of the subspace cluster. $p_{ij,k}^{l(q)}$ is a weight coefficient that indicates the extent to which e_j can gather information about e_i from the k -th aspect. When calculating $p_{ij,k}^{l(q)}$, the L_1 distance is employed as the similarity metric, which is consistent with the process of aligning entities (Section 4.4). To start the iterative clustering, $p_{ij,k}^{l(0)}$ is initialized to $1/K$.

Relational reflective neighborhood routing is driven by the following two plausible hypotheses: (1) For a subset of N_{e_i} , if all entities in this subset are related to the k -th aspect, they will be similar and form a cluster in the k -th subspace; and (2) If entities e_i and e_j are similar in the k -th subspace, the relation connecting them can describe the k -th aspect. Hypothesis (1) necessitates a multi-round clustering process, while hypothesis (2) acts as a strong prior to guide clustering for faster convergence.

After Q-round clustering, the weight of neighbor e_j , i.e., $p_{ij,k}^{l(Q)}$, is obtained, which serves to disentangle the neighborhood of the entity e_i . In addition, to further highlight the significance of e_j for e_i under the relation r_m in the k -th aspect, the relation attention coefficient is incorporated into the final neighbor weights, as follows:

$$\alpha_{ij,k}^l = \frac{\exp(p_{ij,k}^{l(Q)} + (\mathbf{a}^l)^T \mathbf{h}_{r_m,k})}{\sum_{(r_n, e_v) \in N_{e_i}} \exp(p_{iv,k}^{l(Q)} + (\mathbf{a}^l)^T \mathbf{h}_{r_n,k})}, \quad (6)$$

where $\mathbf{a}^l$ is a trainable relation attention weight vector and r_m is the relation connecting entities e_i and e_j . It is also noteworthy that (4) applies *softmax* across different aspects, while (6) applies *softmax* across different neighbors within the same aspect. The dual *softmax* allows for inter-aspect and intra-aspect information interaction.

The hidden representations of entities are updated by aggregating neighboring entities as follows:

$$\mathbf{h}_{e_i,k}^{l+1} = \sum_{(r_m, e_j) \in N_{e_i}} \alpha_{ij,k}^l \mathbf{M}_{r_m,k} \mathbf{h}_{e_j,k}^l, \quad (7)$$

where $\mathbf{h}_{e_i,k}^{l+1}$ is the updated entity representation for e_i in the k -th aspect, serving as input for the $(l+1)$ -th layer. Because entity embeddings at distinct layers encapsulate different alignment information and collectively contribute to the efficacy of EA, it is necessary to integrate the outputs of different layers to yield the following global entity representations for assessing the similarity between entities:

$$\mathbf{h}_{e_i,k}^{\text{global}} = [\mathbf{h}_{e_i,k}^1 \parallel \dots \parallel \mathbf{h}_{e_i,k}^L], \quad (8)$$

where $\parallel$ represents vector concatenation, and L denotes the total number of layers.

4.3 Macro-disentanglement with mutual information regularization

Previous studies (Liu et al. 2020; Zheng et al. 2024) have indicated that, in addition to micro-disentanglement, regularization constraints should be imposed on the node representations from the macro level to achieve comprehensive disentanglement. Drawing on the work of Wu et al. (2021) and Cheng et al. (2020), we choose mutual information regularization, as it can evaluate both linear and nonlinear correlations between random variables and is applicable to multi-relational complex scenarios in KGs. With the constraint, the disentangled aspect representations are encouraged to be independent of each other and contain as much different information as possible.

Specifically, we employ the contrastive log-ratio upper bound as an estimator to calculate the upper bound of mutual information, thus achieving the goal of minimizing mutual information. Since the true conditional probability distribution $p(\mathbf{h}_{e,m}^{\text{global}} | \mathbf{h}_{e,n}^{\text{global}})$ between aspects is unknown, a simple neural network (two fully connected layers followed by an activation layer) with parameter θ is adopted to construct a variational distribution $q_\theta(\mathbf{h}_{e,m}^{\text{global}} | \mathbf{h}_{e,n}^{\text{global}})$. This variational distribution is used to approximate the true conditional distribution. The upper bound of mutual information is thus defined as:

$$L_{ub} = \min_{\theta} \sum_{m=1}^K \sum_{n=1}^K [\log q_\theta(\mathbf{h}_{e,m}^{\text{global}} | \mathbf{h}_{e,n}^{\text{global}}) - \log q_\theta(\mathbf{h}_{e,m}^{\text{global}} | \mathbf{h}_{e',n}^{\text{global}})], \quad (9)$$

where $\log q_\theta(\mathbf{h}_{e,m}^{\text{global}} | \mathbf{h}_{e,n}^{\text{global}})$ and $\log q_\theta(\mathbf{h}_{e,m}^{\text{global}} | \mathbf{h}_{e',n}^{\text{global}})$ denote the log-likelihood of positive and negative sample pairs,

respectively. Furthermore, it is necessary to minimize the KL divergence between the conditional and variational distributions to ensure that the results obtained by (9) remain the mutual information upper bound. Consequently, the final optimization objective for macro-disentanglement can be stated as follows:

$$L_{\text{ma-reg}} = L_{ub} + \min_{\theta} \mathbb{D}_{\text{KL}}[p(\mathbf{h}_{e,m}^{\text{global}} | \mathbf{h}_{e,n}^{\text{global}}) \| q_{\theta}(\mathbf{h}_{e,m}^{\text{global}} | \mathbf{h}_{e,n}^{\text{global}})], \quad (10)$$

where $p(\mathbf{h}_{e,m}^{\text{global}} | \mathbf{h}_{e,n}^{\text{global}})$ is the conditional probability distribution. The regularization loss and the EA loss will be optimized alternately.

4.4 Entity alignment with alignment-aware self-attention

In our method, the representations of entities consist of embeddings in multiple disentangled aspects and the contribution of different aspects varies. Therefore, we first calculate the distance between entities within each aspect. For the entity pair (e_1, e_2) , the distance in k -th aspect is defined as follows:

$$d(e_1, e_2)_k = \|\mathbf{h}_{e_1,k}^{\text{global}} - \mathbf{h}_{e_2,k}^{\text{global}}\|_{L_1}. \quad (11)$$

Subsequently, an alignment-aware self-attention strategy is proposed. It is capable of adaptively assigning appropriate weights to different aspects. The final distance of (e_1, e_2) is calculated as the following weighted sum:

$$d(e_1, e_2) = \sum_{k=1}^K \beta_{12,k} d(e_1, e_2)_k, \quad (12)$$

$$\beta_{12,k} = K \frac{\exp(\mathbf{b}^T [\mathbf{h}_{e_1,k}^{\text{global}} \| \mathbf{h}_{e_2,k}^{\text{global}}])}{\sum_{k'=1}^K \exp(\mathbf{b}^T [\mathbf{h}_{e_1,k'}^{\text{global}} \| \mathbf{h}_{e_2,k'}^{\text{global}}])}, \quad (13)$$

where $\mathbf{b}$ is a trainable attention weight vector.

In training, the embeddings of aligned entity pairs (i.e., positive samples) are expected to be close to each other, while the embeddings of unaligned entity pairs (i.e., negative samples) are expected to be far away from each other. Therefore, the margin-based objective function is defined for EA:

$$L_{ea} = \sum_{(e_i, e_j) \in P} \sum_{(e'_i, e'_j) \in P'} \max\{0, d(e_i, e_j) - d(e'_i, e'_j) + \gamma\}, \quad (14)$$

where P denotes the set of positive samples, P' denotes the set of negative samples, and $\gamma > 0$ is a margin hyper-parameter used to separate positive and negative samples.

Some previous methods generate negative samples by nearest neighbor sampling. Although proven to be effective, it greatly increases the computational cost. To improve efficiency, we obtain negative samples by replacing entities in the pre-aligned entity pairs with randomly selected entities. Losses for macro-disentanglement (10) and EA (14) are optimized in an alternating manner throughout the training process.

Figure 4 illustrates the entity alignment process using alignment-aware self-attention, with K set to 3. Across different aspect spaces, disentangled representations remain independent because of mutual information regularization. In each aspect space, the distances between entity pairs and their corresponding attention weights are computed separately. The results from multiple aspects are then integrated to determine the final distance. These aligned entities serve as anchors, and bringing them closer can enhance the semantic consistency of different KGs within the same aspect.

4.5 Semi-supervised learning with multi-aspect seed generators

Insufficient training data constrains the performance of embedding-based EA methods. Existing methods generate new alignment seeds iteratively to solve this problem. For instance, MMEA (Mao et al. 2020a) proposes a bi-directional iterative strategy: If the embeddings of entities e_i and e_j in two KGs are most similar to each other, then the (e_i, e_j) pair will be identified as a potential alignment seed. On the other hand, in our method RRDGNN, determining the number of entities' aspects with a single value (i.e., K) is not enough. Entities in KGs are not equally informative. Intuitively, a large K is suitable for entities with dense neighborhoods, while entities with sparse neighborhoods require a small K .

In light of the above two considerations, we devise a semi-supervised learning framework in which multiple seed generators with different aspect values are co-trained. The generators employ the bi-directional iterative algorithm to identify new alignment seeds with specific sparsity. This allows different generators to label entities with different sparsity, and their collaboration improves the quality of generated seeds. Specifically, the sparsity of the entity e_i is defined as follows:

$$s_{e_i} = 1 - \frac{|N_{e_i}|}{|N_{e^*}|}, \quad (15)$$

where e^* is the entity with the most neighbors. All entities are ranked by sparsity in descending order and divided into M equal parts. If entity e_i is in the m -th part, its sparsity ranking is as follows:

$$sr_{e_i} = m. \quad (16)$$

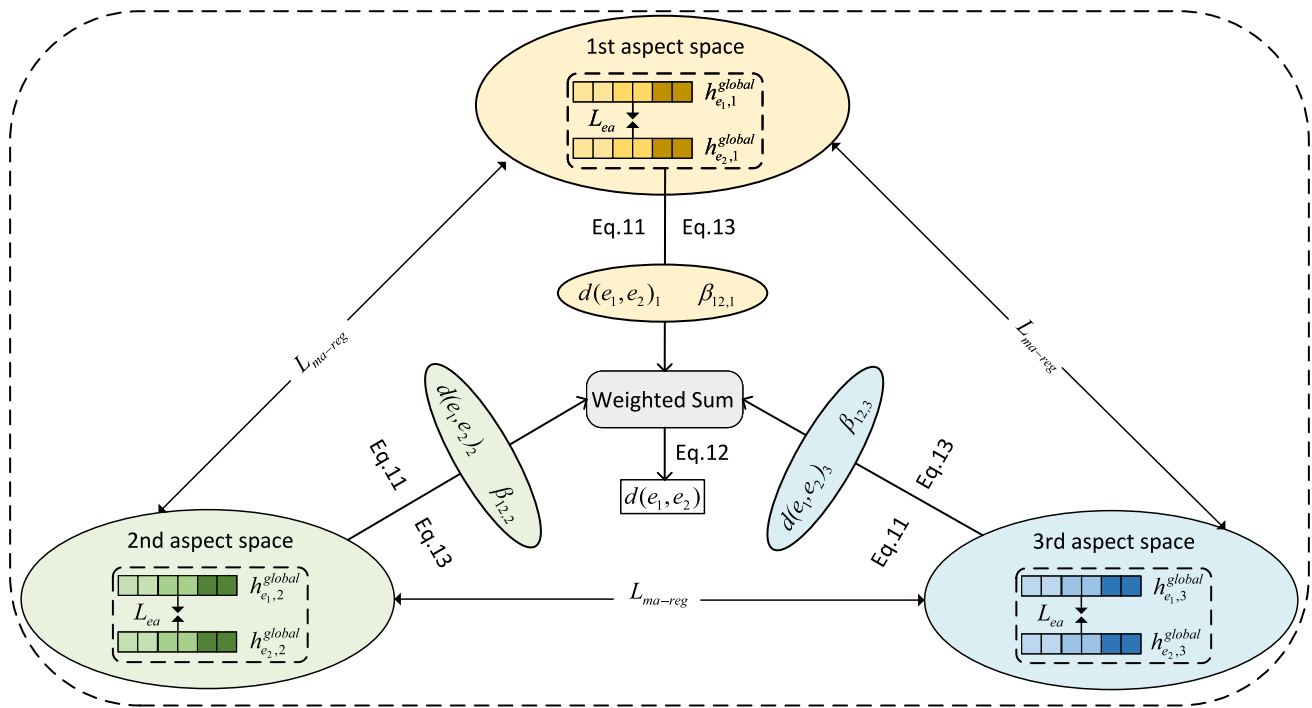


Fig. 4 Case for entity alignment with 3 aspects

Algorithm 1 Semi-supervised learning with multi-aspect seed generators.

Input: KG_1 and KG_2 (knowledge graphs to be aligned), P^0 (initial alignment seeds), M (number of seed generators), $maxIt$ (maximum number of iterations).

Output: $\{\mathbf{H}_{E_1, m}^{maxIt}\}_{m=1}^M$ and $\{\mathbf{H}_{E_2, m}^{maxIt}\}_{m=1}^M$ (final entity embedding matrices).

```

1: Calculate the sparsity ranking of entities using (16);
2: Arrange the  $M$  seed generators in ascending order according to their  $K$  values;
3: for  $t = 1$  to  $maxIt$  do
4:    $Set_{new} = \emptyset$ ;
5:   for  $m = 1$  to  $M$  do
6:     Train the  $m$ -th seed generator a number of epochs with  $P^{t-1}$  using (10) and (14) to get  $\mathbf{H}_{E_1, m}^t$  and  $\mathbf{H}_{E_2, m}^t$ ;
7:     Employ the bi-directional iterative strategy (Mao et al. 2020a) to obtain the new alignment seeds  $Set_m$ ;
8:     for each  $(e_i, e_j) \in Set_m$  do
9:       if  $\lfloor (sr_{rank_{e_i}} + sr_{rank_{e_j}}) \div 2 \rfloor \neq m$  then
10:         $Set_m.pop((e_i, e_j))$  // filter out entity pairs whose sparse ranking is not  $m$ ;
11:       end if
12:     end for
13:    $Set_{new} = Set_{new} \cup Set_m$ ;
14: end for
15:  $P^t = P^{t-1} \cup Set_{new}$ ;
16: end for
17: return  $\{\mathbf{H}_{E_1, m}^{maxIt}\}_{m=1}^M$  and  $\{\mathbf{H}_{E_2, m}^{maxIt}\}_{m=1}^M$ .

```

Correspondingly, M seed generators are sorted in ascending order according to their K values. The m -th seed generator can only select entity pairs with sparsity ranking

equal to m , and seeds with other sparsities are filtered out. The outputs of all seed generators are merged and added to the training set of the previous iteration. The specific process of semi-supervised learning with multi-aspect seed generators is illustrated in Algorithm 1.

5 Experiments

5.1 Experimental settings

Datasets The RRDGNN model is evaluated on the following two widely used real-world EA datasets:

- DBP15K (Sun et al. 2017) is built from the multilingual Depedia, which comprises three cross-lingual KG pairs: DBP_{ZH-EN} (Chinese-English), DBP_{JA-EN} (Japanese-English), and DBP_{FR-EN} (French-English).
- S-DBP15K (Guo et al. 2019) comprises two cross-lingual subsets: S-DBP_{FR-EN} (French-English) and S-DBP_{DE-EN} (German-English). The entities' neighborhoods in S-DBP15K are more sparse, thereby presenting a greater challenge for entity alignment.

Each of the five subsets contains 15,000 inter-language links (ILLs) from English entities to their counterparts in other languages. These ILLs can be considered gold standards. Following previous studies, 30% of the ILLs are

allocated for training, while the remaining 70% are set aside for testing. Table 2 provides the detailed statistics of the datasets. For clarity, Ent., Rel., Tri., and Pairs represent entities, relations, relation triples, and pre-aligned entity pairs, respectively.

Baselines In the main experiment, RRDGNN is evaluated in comparison with 18 recent EA methods. They are 5 translation-based methods: MTransE (Chen et al. 2017), JAPE (Sun et al. 2017), BootEA (Sun et al. 2018), TransEdge (Sun et al. 2019b), and ESEA (Jiang et al. 2022), 11 GNN-based methods: GCN-Align (Wang et al. 2018), AliNet (Sun et al. 2020), MRAEA (Mao et al. 2020a), Dual-AMN (Mao et al. 2021), DuGa-DIT (Xie et al. 2021), GALA (Zhang et al. 2023), DSEA (Qian and Pan 2023a), BNGNN (Shi et al. 2023), RHGNN (Zhang et al. 2024), SCMEA (Zhou et al. 2024) and SPSEA (Yan et al. 2024), 2 methods using other encoding models: RpAlign (Huang et al. 2022) and LightEA (Mao et al. 2022). RpAlign converts entity alignment into the KG completion problem and uses pRotatE (Sun et al. 2019a) for knowledge embedding. LightEA applies the label propagation algorithm to KG representation learning through the three-view label propagation mechanism. Furthermore, RRDGNN is compared with 10 recent methods: DuGa-DIT (Xie et al. 2021), RSA (Akhtar et al. 2022), EMGCN (Tam et al. 2022), DSEA (Qian and Pan 2023a), BNGNN (Shi et al. 2023), RHGNN (Zhang et al. 2024), SCMEA (Zhou et al. 2024), HOLI-GNN (Chen et al. 2024), AKR (Akhtar et al. 2025) and SCSA (Li et al. 2025) in a setting where pre-trained word embeddings are used.

Hyperparameter settings A reasonable range is first determined for each hyperparameter, and the optimal value is selected based on experimental results. Although this approach may only lead the model to a local rather than a global optimum, it still yields highly competitive performance. Specifically, the following hyperparameter search

spaces are considered: the temperature hyperparameter τ in (4): {0.005, 0.01, 0.05, 0.1, 0.5}, the margin hyperparameter γ in (14): {1.0, 1.3, 1.5, 1.7, 2.0}, the maximum number of iterations for micro-disentanglement clustering (i.e., Q): {1, 3, 5, 7, 9}, the dimension of entity and relation embeddings for each aspect: {16, 32, 64, 96, 128}, the number of aggregation layers in RRDGNN: {1, 2, 3, 4}, and the number of negative samples for each entity in (14): {16, 32, 64, 96, 128}. The RMSprop optimizer is employed for training, with a learning rate selected from {0.0005, 0.001, 0.005} and a dropout rate from {0.2, 0.3, 0.4}.

The final hyperparameters used for RRDGNN are as follows: $\tau = 0.01$, $\gamma = 1.3$, $Q = 3$, the embedding dimension is 64, the number of RRDGNN layers is 2, the number of negative samples is 32, the learning rate is 0.005, and the dropout rate is 0.3. In the model variant with a single seed generator, the number of aspects is set to $K = 5$. In the setting with multiple seed generators, two additional generators with $K = 3$ and $K = 7$ are trained simultaneously. The model is trained for 500 epochs during basic training, followed by 600 additional epochs for iterative learning.

Implementation Our model is implemented using TensorFlow. All experiments are conducted on a server powered with an Ubuntu 20.04 system, an Intel Xeon Platinum 8358P CPU, and an NVIDIA Tesla A40 (48GB) GPU. To obtain stable experimental results, the code is run five times independently and the mean is calculated.

Model size The total number of trainable parameters for our model is $(|E_1| + |E_2| + |R_1| + |R_2|)Kd + (K + 2)Ld$, where L is the number of RRDGNN layers, K is the number of aspects, and d is the dimension of entity and relation embeddings for each aspect. With the above optimal hyperparameters, the average number of parameters is 14.23M for the three subsets of dataset DBP15K, and 9.84M for the two subsets of dataset S-DBP15K.

Evaluation metrics To assess the efficacy of our method, we utilize two key metrics: Hits@k and *mean reciprocal rank* (MRR). Higher Hits@k and MRR values signify better performance.

5.2 Main results

5.2.1 Results on DBP15K

Table 3 illustrates the performance comparison of all methods on DBP15K. The majority of the results are drawn from the original papers, while the rest are derived from references Sun et al. (2020) and Mao et al. (2022). As evidenced in the table, the following observations can be made:

Table 2 Detailed information of DBP15K and S-DBP15K

Datasets		Ent.	Rel.	Tri.	Pairs
DBP _{ZH-EN}	ZH	19,388	1,701	70,414	15,000
	EN	19,572	1,323	95,142	
DBP _{JA-EN}	JA	19,814	1,299	77,214	15,000
	EN	19,780	1,153	93,484	
DBP _{FR-EN}	FR	19,661	903	105,998	15,000
	EN	19,993	1,208	115,722	
S-DBP _{FR-EN}	FR	15,000	177	33,532	15,000
	EN	15,000	221	36,508	
S-DBP _{DE-EN}	DE	15,000	120	37,377	15,000
	EN	15,000	222	38,363	

Table 3 Experimental results on DBP15K

Methods	DBP _{ZH-EN}			DBP _{JA-EN}			DBP _{FR-EN}		
	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR
MTransE	0.308	0.614	0.364	0.279	0.575	0.349	0.244	0.556	0.335
JAPE	0.412	0.745	0.490	0.363	0.685	0.476	0.324	0.667	0.430
BootEA	0.629	0.847	0.703	0.622	0.854	0.701	0.653	0.874	0.731
TransEdge	0.735	0.919	0.801	0.719	0.932	0.795	0.710	0.941	0.796
ESEA	0.725	0.869	0.775	0.750	0.878	0.795	0.794	0.909	0.834
GCN-Align	0.413	0.744	0.549	0.399	0.745	0.546	0.373	0.745	0.532
AliNet	0.539	0.826	0.628	0.549	0.831	0.645	0.552	0.852	0.657
MRAEA	0.757	0.930	0.827	0.758	0.934	0.826	0.781	0.948	0.849
Dual-AMN	0.808	0.940	0.857	0.801	0.949	0.855	0.840	0.965	0.888
DuGa-DIT	0.756	0.858	0.805	0.781	0.914	0.829	0.837	0.920	0.867
GALA	0.570	-	0.650	0.585	-	0.660	0.607	-	0.690
DSEA	0.681	0.816	0.725	0.706	0.842	0.752	0.730	0.896	0.786
BNGNN	0.792	<u>0.949</u>	0.851	0.779	<u>0.955</u>	0.844	0.793	<u>0.966</u>	0.858
RHGNN	0.694	0.889	0.766	0.761	0.921	0.821	0.768	0.951	0.839
SCMEA	0.747	0.933	0.816	0.775	0.940	0.838	0.765	0.950	0.835
SPSEA	<u>0.821</u>	0.943	<u>0.859</u>	0.817	0.938	<u>0.867</u>	0.852	0.949	0.867
RpAlign	0.748	0.889	0.794	0.730	0.890	0.782	0.752	0.900	0.801
LightEA	0.812	0.915	0.849	<u>0.821</u>	0.933	0.864	<u>0.863</u>	0.959	<u>0.900</u>
RRDGNN	0.851	0.964	0.893	0.839	0.969	0.888	0.864	0.979	0.907

The optimal results are indicated by bold numerals, while the second-best results are underlined

- In comparison to translation-based methods, RRDGNN demonstrates notable advantages across all datasets. The Hits@1 and Hits@10 of RRDGNN are superior to those of the baseline methods by a minimum of 7% and 3.7%, respectively. This illustrates that GNN is a more expressive knowledge embedding model than TransE. GNN is capable of aggregating local neighborhood information to capture the similarity of aligned entities, whereas TransE is constrained by its inability to handle complex relations.
- Among the GNN-based baseline methods, SPSEA and BNGNN rank top 2. The former performs optimally on metrics Hits@1 and MRR due to its high-precision seed labeling strategy. The latter performs optimally on Hits@10, because it efficiently fuses the alignment information in the entity attributes, thereby yielding superior results at a coarse-grained level. Nevertheless, RRDGNN still demonstrates better performance compared to the two methods across all metrics. For example, the Hits@1 of RRDGNN is 2.1% superior to that of the method SPSEA on average on three sub-datasets. This is because RRDGNN can address the inherent limitations of most existing GNN-based methods by implementing disentangled entity representations.
- Compared with methods that do not use TransE or GNN for encoding, RRDGNN continues to demonstrate excellent performance. Although RRDGNN only exhibits

slight superiority over LightEA on the DBP_{FR-EN} dataset, it performs much better than LightEA on DBP_{ZH-EN} and DBP_{JA-EN}. For instance, the Hits@1 of RRDGNN is nearly 4% higher than that of LightEA on DBP_{ZH-EN}.

5.2.2 Results on S-DBP15K

Long-tail entities in KGs refer to entities with small degrees. They are difficult to align because they can hardly obtain useful information from neighboring entities. RRDGNN is compared with multiple competitive baseline methods on S-DBP15K to evaluate its effectiveness for aligning long-tail entities. The overall results are shown in Table 4.

It is clear that the performances of all methods decline considerably on all metrics. This can be ascribed to the fact that the sparse neighborhood structure cannot provide sufficient alignment information. Furthermore, when comparing the experimental results on datasets DBP_{FR-EN} and S-DBP_{FR-EN}, it can be observed the performance degradation of LightEA is more obvious than that of RRDGNN. For example, the Hits@1 of LightEA on DBP_{FR-EN} is only 0.1% worse than our method, while on S-DBP_{FR-EN}, the gap reaches 2.3%. This shows the good robustness of RRDGNN. Overall, our method surpasses other approaches on most metrics, even in sparse scenarios. The potential

Table 4 Experimental results on S-DBP15K datasets

Methods	S-DBP _{FR-EN}			S-DBP _{DE-EN}		
	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR
BootEA	0.365	0.649	0.460	0.503	0.732	0.580
TransEdge	0.400	0.675	0.490	0.556	0.753	0.630
GCN-Align	0.270	0.559	0.367	0.404	0.651	0.490
Dual-AMN	0.481	<u>0.778</u>	0.568	0.614	0.823	0.687
LightEA	0.484	0.769	0.570	0.615	0.817	0.685
GALA	0.295	-	0.360	0.431	-	0.500
BNGNN	0.486	0.775	0.584	0.597	0.824	0.677
SCMEA	<u>0.497</u>	0.759	<u>0.587</u>	0.616	0.831	0.692
RRDGNN	0.507	0.779	0.596	0.616	<u>0.826</u>	<u>0.688</u>

Bold represents the highest value and the underline the second highest

reason is that RRDGNN can learn disentangled entity representations and assign different weights to different aspects.

5.3 Using pre-trained word embeddings

Some studies have indicated that the alignment results can be markedly enhanced by integrating pre-trained word embeddings of entity names. This is because pre-trained word embeddings contain textual features of entities, which are complementary to structural features. In this subsection, we implement a model variant n-RRDGNN based on pre-trained word embeddings of entity names, so that our method can be fairly compared with these approaches.

Referring to previous work (Shi et al. 2023), the pre-trained multilingual BERT (Devlin et al. 2019) is utilized to generate word embeddings for entity names. Specifically, for entity e_i , its name is represented as $MLP(CLS(name_{e_i}))$, where $CLS(name_{e_i})$ denotes the embedding of the token CLS, and MLP denotes an MLP layer. BERT is fine-tuned

using (14) as the objective function, and its parameters will be frozen when combined with RRDGNN. The final distance matrix between entities is $D = (normalization(D_n) + normalization(D_s))/2$, where D_n is the textual distance matrix calculated with pre-trained word embeddings and D_s is the structural distance matrix calculated with entity embeddings in RRDGNN. In semi-supervised learning, multiple seed generators select newly aligned entity pairs based on the matrix D .

Experimental results of the methods utilizing entity names on DBP15K are presented in Table 5. It can be observed that after integrating pre-trained word embeddings, RRDGNN's performance has been significantly improved. For example, the Hits@1 exhibits an increase of 11.6%, 14.6%, and 13.2% on the three sub-datasets, respectively. In addition, our method achieves better performance than other approaches, particularly on DBP_{ZH-EN}, where Hits@1 surpasses the best baseline by 1.3%. Regardless of whether pre-trained word embeddings are used or not, our method demonstrates excellent performance on all datasets and metrics.

Table 5 Performance of methods using entity names on DBP15K datasets

Methods	DBP _{ZH-EN}			DBP _{JA-EN}			DBP _{FR-EN}		
	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR
DuGa-DIT	0.807	0.882	0.832	0.914	0.952	0.928	0.982	0.992	0.985
RSA	0.755	0.889	-	0.802	0.939	-	0.914	0.974	-
EMGCN	0.863	0.946	-	0.866	0.952	-	0.940	0.989	-
DSEA	0.789	0.892	0.819	0.791	0.894	0.826	0.847	0.951	0.883
BNGNN	0.946	<u>0.989</u>	<u>0.964</u>	0.967	0.996	0.979	0.989	0.999	0.994
RHGNN	0.768	0.963	0.843	0.799	0.972	0.864	0.939	0.992	0.959
SCMEA	0.908	0.979	0.935	0.934	0.987	0.954	0.982	0.998	0.988
HOLI-GNN	0.901	0.966	0.926	0.924	0.977	0.943	0.971	0.993	0.980
AKR	0.794	0.929	-	0.813	0.962	-	0.938	0.981	-
SCSA	<u>0.954</u>	0.981	<u>0.964</u>	0.990	0.998	0.993	<u>0.995</u>	0.998	<u>0.996</u>
n-RRDGNN	0.967	0.995	0.979	<u>0.985</u>	0.998	<u>0.991</u>	0.996	0.999	0.998

Bold represents the highest value and the underline the second highest

5.4 Ablation study

To investigate the effectiveness of different components in RRDGNN, the following five variants have been devised to conduct ablation studies:

RRDGNN-single: This is a variant model that just uses a single seed generator.

w/o Iter: This is a basic version that does not iteratively generate new alignment seeds.

w/o DisEnt: The disentanglement process is removed from RRDGNN-single, reducing the model to represent the entity with a single embedding. The dimension of the single embedding is identical to that of the concatenated multiple-aspect embeddings.

w/o MI: The macro-disentanglement constraint based on mutual information regularization is removed from RRDGNN-single.

w/o Attn: The alignment-aware self-attention is removed from RRDGNN-single. All aspects are assigned the same weight in this variant.

We removed these components from RRDGNN-single instead of the full RRDGNN in order to prevent the refinement process with multi-aspect seed generators from interfering with the results of ablation studies. Experimental results of this part are presented in Table 6.

Effect of multi-aspect seed generators Multiple seed generators with different aspect values are trained simultaneously to select new entity pair seeds with specific sparsity. The results of the variant model RRDGNN-single and RRDGNN in Table 6 demonstrate that this strategy can bring a considerable improvement in EA performance, particularly for metrics Hits@1 and MRR. As an illustration, on the DBP_{JA-EN} dataset, the Hits@1 and MRR can be enhanced by 2.1% and 1.5%, respectively. This is because the different seed generators have a preference for entities with specific sparsity, and they can collaborate to select high-quality alignment seeds.

Effect of semi-supervised iteration We employ a semi-supervised learning strategy and iteratively select potentially

aligned entity pairs. A comparison of the results of variants RRDGNN-single and w/o Iter in Table 6 reveals that the iteration process is indeed effective. Removing the semi-supervised iteration results in a notable decline in model performance. To illustrate, Hits@1 drops by at least 4.8% on DBP15K and 1.8% on S-DBP15K.

Effect of disentangled representations We learn disentangled entity representations at both the micro and macro levels. To further evaluate the effectiveness of this strategy, we develop a variant w/o DisEnt, which omits micro and macro disentanglement. The results in Table 6 indicate that this variant performs poorly. Compared to RRDGNN-single, Hits@1 of the variant decreases by an average of 3.4% on DBP15K. The performance degradation is more pronounced on S-DBP15K, with Hits@1 decreasing by an average of 4.4%. This demonstrates that multiple disentangled representations are better than a single embedding, as they can capture distinct information regarding different aspects of the entity.

Effect of the macro-disentanglement constraint To reduce the correlation between embeddings of different aspects, the mutual information regularization is applied to the final disentangled entity representations. w/o MI in Table 6 denotes a variant model that does not include mutual information regularization. By comparing the results of w/o MI and RRDGNN-single, the macro-disentanglement constraint proves to be useful. Removing this component leads to a decline of more than 1% in Hits@1 on the DBP15K datasets.

Effect of the alignment-aware self-attention The alignment-aware self-attention can adaptively assign appropriate weights to different aspects when computing the similarity of entity pairs. w/o Attn in Table 6 represents a variant model that does not utilize the aforementioned attention mechanism. A comparison of the results of RRDGNN-single and w/o Attn demonstrates that removing this component results in a

Table 6 Ablation studies on DBP15K and S-DBP15K

Methods	DBP_{ZH-EN}			DBP_{JA-EN}			DBP_{FR-EN}			$S-DBP_{FR-EN}$			$S-DBP_{DE-EN}$		
	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR
RRDGNN-single	0.833	0.963	0.882	0.818	0.964	0.873	0.847	0.977	0.897	0.491	0.777	0.585	0.603	0.819	0.679
w/o Iter	0.777	0.948	0.839	0.768	0.954	0.836	0.799	0.967	0.861	0.470	0.758	0.566	0.585	0.814	0.664
w/o DisEnt	0.790	0.937	0.845	0.779	0.943	0.839	0.826	0.965	0.878	0.444	0.713	0.533	0.563	0.772	0.635
w/o MI	0.822	0.960	0.874	0.807	0.961	0.865	0.833	0.972	0.886	0.487	0.775	0.582	0.600	0.818	0.678
w/o Attn	0.826	0.958	0.876	0.815	0.963	0.870	0.841	0.975	0.893	0.481	0.763	0.573	0.598	0.806	0.671
RRDGNN	0.851	0.964	0.893	0.839	0.969	0.888	0.864	0.979	0.907	0.507	0.779	0.596	0.616	0.826	0.688

Bold represents the highest value and the underline the second highest

decline in model performance. It is noteworthy that the performance degradation is more pronounced on S-DBP15K than on DBP15K. This indicates that the strategy may play a greater role in sparse EA scenarios.

5.5 Further analysis

Robustness to neighborhood heterogeneity between KGs To better verify that RRDGNN can handle the problem of neighborhood heterogeneity, we have constructed a more challenging EA scenario based on DBP_{ZH-EN} , where the heterogeneity between KGs is more serious. Specifically, a portion of triples is randomly removed from the Chinese KG, while the English KG remains unchanged. This procedure will improve the structural heterogeneity between the two KGs. The percentage of removed triples ranges from 10% to 50%, with a step of 10%. RRDGNN is compared with two competitive methods, Dual-AMN and LightEA, which only leverage the structural information in KGs. The experimental results are presented in Table 7. From the table, it can be found that:

When the number of removed triples increases, the results of all methods decrease significantly. This indicates that neighborhood heterogeneity is a key factor limiting the performance of EA methods. However, RRDGNN is consistently superior to other compared methods. Furthermore, as heterogeneity rises, the advantage of our method becomes more evident. For example, when only 10% of triples are removed, Hits@1 of RRDGNN is relatively 5.9% higher than Dual-AMN. Upon this ratio reaching 50%, RRDGNN's relative advantage on Hits@1 reaches 6.2%. This phenomenon is more pronounced on the metric Hits@10, where the relative advantage of RRDGNN increases from 3.8% to 7.3%. These observations present good robustness of RRDGNN to neighborhood heterogeneity between KGs.

Sensitivity to the number of pre-aligned entity pairs To verify whether our method is sensitive to the number of pre-aligned entity pairs, we compare RRDGNN with two

other GNN-based methods, Dual-AMN and BNGNN, on DBP15K datasets. Different proportions of pre-aligned entity pairs are used as training data, ranging from 20% to 50%, with an increase of 10%. The results are shown in Fig. 5. As the number of alignment seeds increases, the performance of all methods improves significantly because they can obtain more supervision signals. Regardless of the number of pre-aligned entity pairs used for training, RRDGNN consistently outperforms compared methods. In addition, compared with the other two methods, the line representing RRDGNN changes more smoothly. Even using only 20% of the pre-aligned entity pairs, the Hits@1 of RRDGNN exceeds 80%. The strong robustness of RRDGNN can be attributed to the co-training of multiple seed generators. These generators collaborate to refine the seed selection process and generate high-quality alignment seeds.

Sensitivity to the number of aspects To analyze the impact of the aspect number on model performance, we consider varying the number of disentangled aspects. The optimal number of disentangled aspects is contingent upon the specific facts in the real datasets. In this study, we use the variant RRDGNN w/o Iter and investigate the aspect numbers within the range of {2, 3, 4, 5, 6, 7, 8}. Figure 6 presents the experimental results on the DBP15K datasets. It can be observed that:

When the aspect number is 2, RRDGNN exhibits suboptimal performance on all datasets. Our inference is that a small value lacks the ability to mine latent aspects and is difficult to handle complex semantics. When the aspect number is increased, the performance on all datasets is improved. The optimal choice appears to be 5 or 6, which is related to the topic number of most entities. When the aspect number is 8, the performance on DBP_{JA-EN} and DBP_{FR-EN} decreases slightly. This is because the division of entity aspects is too fine-grained and cannot capture essential information.

Analysis of the seed degree for different generators To analyze the differences in the sparsity of seeds selected by different generators, we calculate the average degree of new

Table 7 Experimental results of varying degrees of neighborhood heterogeneity on DBP_{ZH-EN}

Methods	Dual-AMN			LightEA			RRDGNN		
	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR
10%	0.772	0.917	0.825	0.777	0.902	0.823	0.818	0.952	0.867
20%	0.731	0.892	0.789	0.742	0.876	0.792	0.780	0.934	0.835
30%	0.681	0.862	0.745	0.693	0.844	0.749	0.732	0.913	0.797
40%	0.616	0.820	0.689	0.632	0.801	0.693	0.668	0.873	0.739
50%	0.544	0.753	0.618	0.551	0.741	0.619	0.578	0.808	0.655

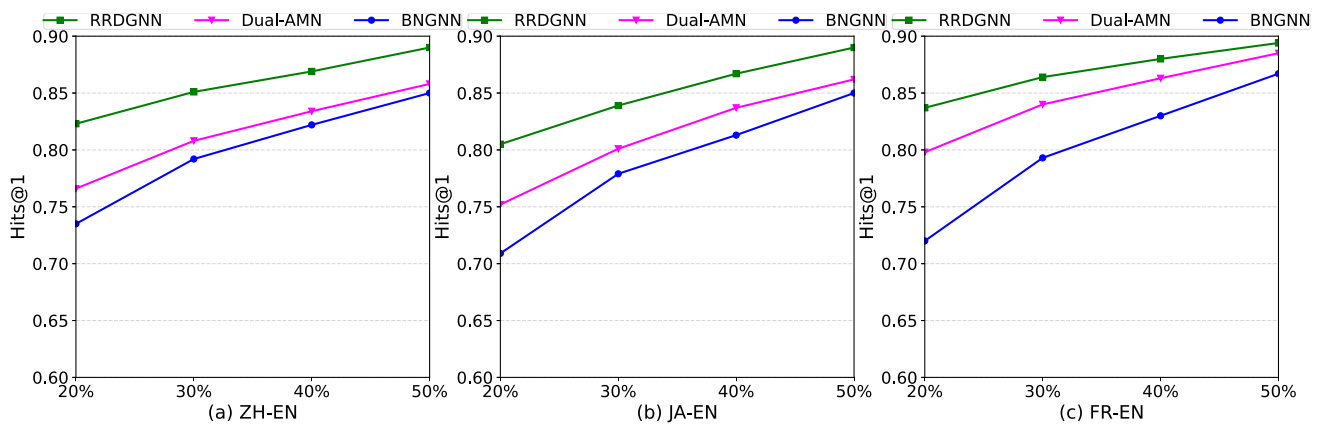


Fig. 5 (a)-(c) show the Hits@1 of RRDGNN, Dual-AMN, and BNGNN using different numbers of pre-aligned entity pairs

seeds across different iterations on DBP_{ZH-EN} . Three seed generators are evaluated with aspect values of 3, 6, and 9, and the number of iterations varies from 1 to 5. The experimental results are presented in Fig. 7. The figure shows that during the first two rounds of iterations, there is no significant difference in the degree of seeds selected by various generators. This occurs because these predicted seeds possess rich structural information, making them easy to label. Starting from the third round, the generators encounter entities with sparse neighborhoods, which are more challenging to align. At this point, the outputs of different generators diverge: those with larger aspect values tend to select entity pairs with higher degrees, while those with smaller aspect values prefer entity pairs with lower degrees. This also validates our hypothesis that different generators can select entities with varying levels of sparsity.

Case study In addition to excellent performance and robustness, our method can also provide certain explainability. We take the entity *Anhui* in the dataset DBP_{ZH-EN} as an example to illustrate the alignment process of RRDGNN. In this case, the number of entity aspects is 3. The visualization result is shown in Fig. 8.

Through the neighbors' attention within a single KG, we can find that the representation of the entity is disentangled. In the Chinese KG (the upper part of Fig. 8), the three aspects mainly focus on people, dialects, and locations. In the English KG, the topics of the second and third aspects are also about dialects and locations. However, the neighbors with high attention in the first aspect are some public facilities, which may be due to the lack of people information related to *Anhui* in the English KG. It can also be observed that the attention of the entity *Hefei* ranks high among different aspects, which

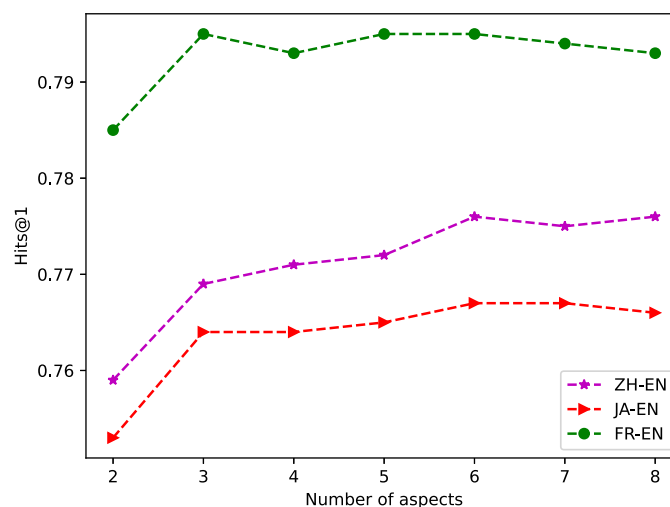


Fig. 6 Impact of the aspect number (K)

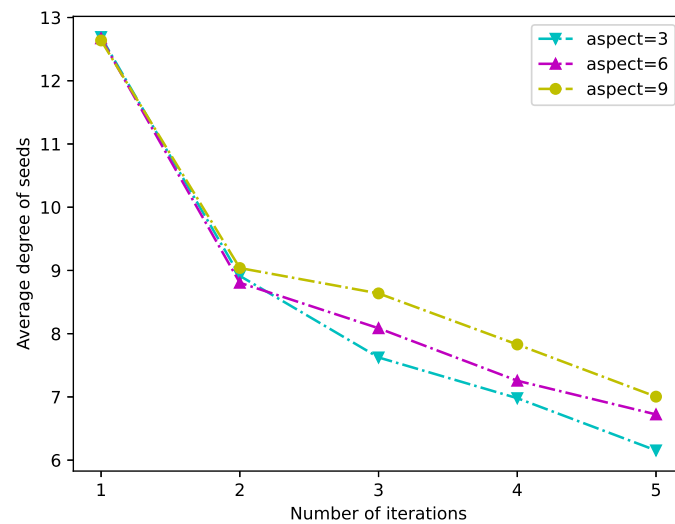
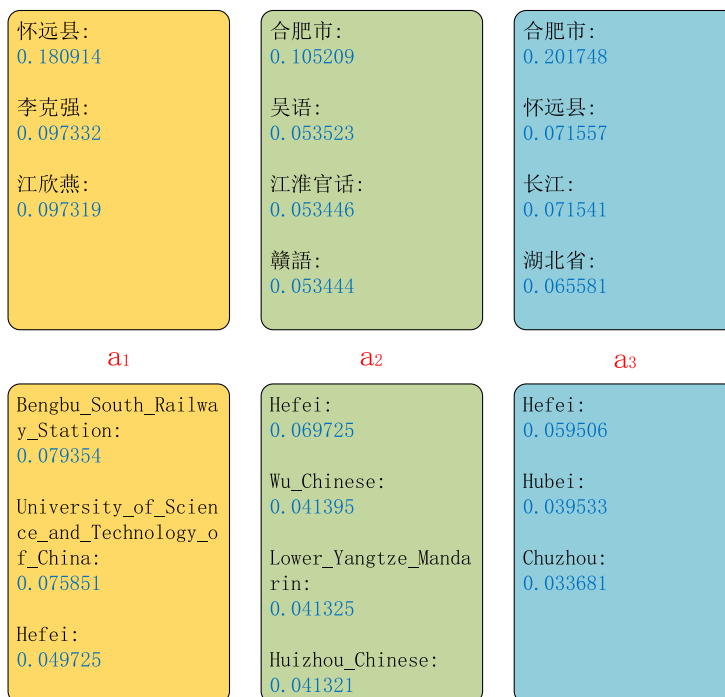


Fig. 7 Average degree of seeds selected by seed generators with different aspect values on DBP_{ZH-EN}

provides key clues for alignment. This can be attributed to the fact that the relation attention coefficient is integrated into the final neighbor weights (6). In addition, the cross-KG attention assigns appropriate weights to different aspects, as

depicted in the right half of Fig. 8. The first pair of aspects has the smallest weight since they focus on different content. The third pair of aspects is given the largest weight because it can provide the most alignment information.

Neighbors with high attention in each aspect.



Neighbors with high attention in each aspect.

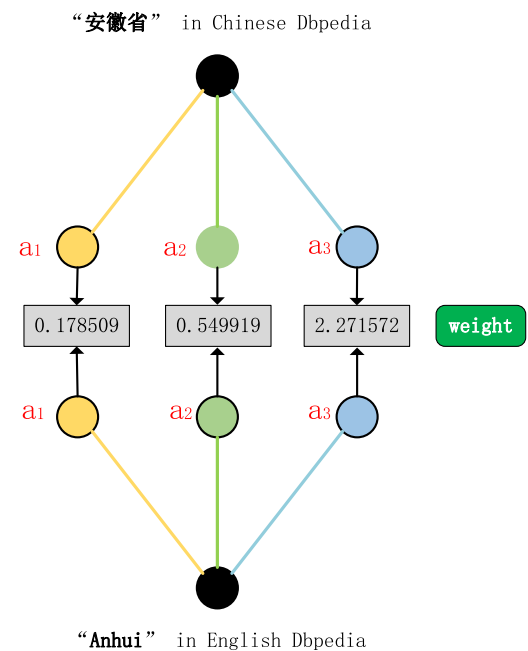


Fig. 8 Case study for the entity *Anhui* on DBP_{ZH-EN} . a_1, a_2, a_3 represent three different aspects

6 Conclusion

In this paper, we examine the problem of entity alignment. We point out the limitations of existing GNN-based EA methods in addressing the neighborhood heterogeneity of equivalent entities. To this end, we propose a novel EA method, RRDGNN. It devises a relational reflective neighborhood routing mechanism and learns disentangled entity representations from both micro and macro levels. Additionally, we design an alignment-aware self-attention strategy that can adaptively assign appropriate weights to different aspects. Finally, we propose to co-train multi-aspect seed generators in a semi-supervised learning framework. They can collaborate to refine the seed selection process and generate high-quality alignment seeds. Experimental results indicate that RRDGNN exhibits obvious superiority over the compared methods on diverse datasets and metrics.

In the future, we will try to design more effective macro-disentanglement algorithms and provide a deeper explanation for the learned disentangled entity representations. Furthermore, we aim to adapt the model for application in other tasks, such as KG-based question-answering and recommender systems.

Acknowledgements This research was supported by the National Natural Science Foundation of China under Grant No. 61972335 and the Postgraduate Research and Practice Innovation Program of Jiangsu Province under Grant No. KYCX24_3738.

Author Contributions Xinchun Shi: Conceptualization, Methodology, Software, Writing - original draft. Bin Li: Conceptualization, Supervision, Funding acquisition. Ling Chen: Validation, Writing - review & editing. Xiaowei Zhang: Investigation, Visualization.

Funding National Natural Science Foundation of China under Grant No. 61972335 and the Postgraduate Research and Practice Innovation Program of Jiangsu Province under Grant No. KYCX24_3738.

Data Availability Data will be made available on request.

Declarations

Conflict of Interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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